

Stat 3113: Lesson 3

Introductory Material Part 3

Project: Experiment Examples



Levels
↓

Example 1: 3^2 ← #Factors

Factorial Experiment

≥ 2 Factors = Explanatory variables

- **Response Variable:** Time for a steel ball to go through fluid from top to bottom of container (units seconds measured on stopwatch)
- **Factor 1:** Diameter of steel ball
- **Levels of Factor 1:** 3 mm, 6 mm, 12 mm
- **Factor 2:** Fluid
- **Levels of Factor 2:** Water, Vegetable Oil, Laundry Detergent

runs / replicates is the same
⇒ balanced Experiment.

Treatments:

||
Specific experimental conditions

Treatment	Diameter	Liquid
1	3mm	water
2	3mm	veg oil
3	3mm	detergent
4	6mm	water
5	6mm	veg oil
6	6mm	detergent
7	12mm	water
8	12mm	veg oil
9	12mm	detergent

Experimental Process:

(To prevent manufacturing defect bias)

*How many times do you want to apply a treatment?
when we apply a treatment
↑ to subject.*

- C* ☐ Constants: There will be one steel ball used of each size for all runs of the experiment. The same type of each individual liquid will be used (same water type, vegetable oil type, and detergent).
- R* ☐ The treatments will be typed into JMP and JMP will randomize the experiment order.
- R* Each treatment will be replicated 4 times for a total of 36 observations. $= 4 \times 9$
- ☐ Each experiment run will consist of filling a container to a set volume with the liquid. The steel ball will be dropped at the center of the container by person 1 after counting down "3, 2, 1, drop". Person 2 will start the stopwatch as the ball is dropped. When the ball hits the bottom of the container, the stopwatch is stopped, and the time (in seconds) is recorded.

→ To be shown in JMP.

*↳ Clearly explain the course of the experiment. ↳ Apply Treatments
↳ Measure the response.*

Experimental Process:

To prevent liquids from mixing $\Rightarrow$ May affect density $\Rightarrow$ Introduce bias!

- ❑ The container will then be cleaned with soap and water followed by rubbing alcohol. The container will be fully dried. Then, the next treatment in the experiment will commence. This will be repeated until all 36 treatments are run.

L_> Inter-trials precautions $\Rightarrow$ To reduce bias as much as possible.

Example 2: 2^3 # Levels # Factors $\Rightarrow$ 3 factors, each with 2 levels.

Factorial Experiment

- **Response Variable:** Time for 4 cups of water to boil (in seconds)

Important to specify the units considered here!

- **Factor 1:** pot used
- **Levels of Factor 1:** large pot, small pot
- **Factor 2:** amount of salt added
- **Levels of Factor 2:** 1 tablespoon, none
- **Factor 3:** stove temperature setting
- **Levels of Factor 3:** high, medium





Treatments:

Treatment	Pot	Salt	Temp
1	large	none	high
2	large	none	low
3	large	1 tbsp	high
4	large	1 tbsp	low
5	small	none	high
6	small	none	low
7	small	1 tbsp	high
8	small	1 tbsp	low

Experimental Process:

- C ☐ Constants: The starting water temperature will ^{be} the same for each trial. The next trial will not start until the stove top ^x is room temperature.
↳ A bias is likely to be introduced between two successive trials!
- R ☐ The treatments will be typed into JMP and JMP will randomize the experiment order. Each treatment will be replicated 4 times for a total of R 32 observations. $= 8 \times 4$
- ☐ Each experiment run will consist of measuring out 4 cups of water and pouring it into the pot. If salt is to be added for the treatment, it is added. The stove will be turned on to the correct temperature setting for the trial by person 1 after a “3, 2, 1, start” countdown.

Experimental Process:

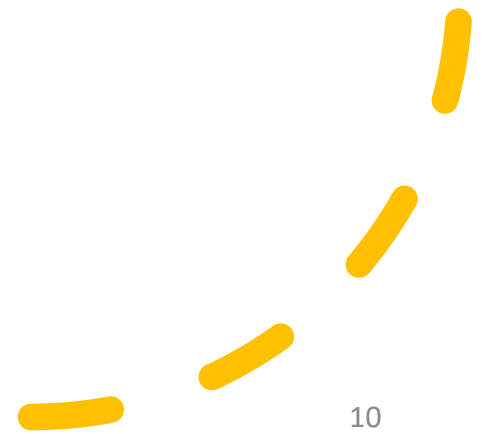
- ❑ As it is being turned on, person 2 will start the stopwatch. The stopwatch will be stopped at the first sign of a clear boil (not a rolling boil, just a simple boil). *↳ Be as specific as possible!*
- ❑ The pot will then be cleaned with soap and water. The pot will be fully dried. Then, the next treatment in the experiment will commence. This will be repeated until all 32 treatments are run.

↳ Inter-trials precautions => To reduce bias as much as possible.

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Reading

Chapter 9 – Section 9.5





Next Topic



Distributions of Quantitative Variables



Chapter 1 - Sections 1.2, 1.3